Praxis

Hi-fi wireframes and information architecture

Version 1.1 · 15 September 2026 · Team Praxis · OpenCV AI Competition 2026 (powered by AWS)
Scope: GRASP upper-limb practice with three camera-coached tasks and an automated Nine-Hole Peg Test
Desktop / landscape tablet, 1280 × 800. Camera panels are stylized placeholders, not renderings of the real overlay.

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2 Information architecture
3 Avatar and reference-panel contract
4 Screens 01-11 (14 frames: light and dark Home, three live-session variants)
5 Design rules applied to every screen

1 Exercise categorization

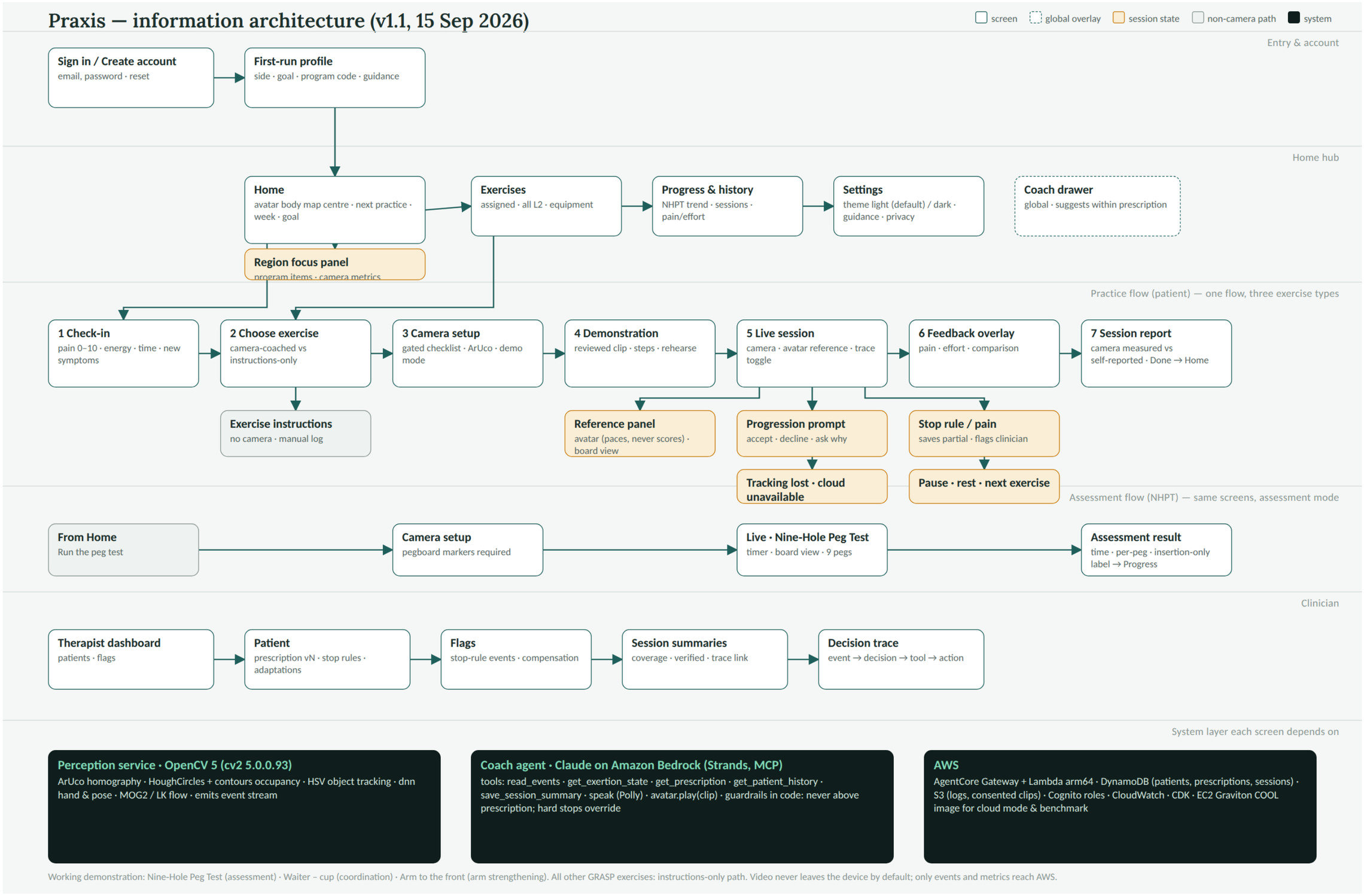
GRASP manual sections plus one added category for the standardized assessment. Three working tasks from three sections; everything else stays instructions-only.

Category	Working (camera-coached)	Pipeline it proves	Non-working (instructions-only)
Standardized assessment	Nine-Hole Peg Test - priority; separate assessment mode	ArUco homography · HoughCircles + contour occupancy · FSM timing	-
Coordination	Waiter - cup (L2, target sheet with 5 dots)	Object-over-target state machine · hand-clear check · reach time	Pouring · ball rolling · drop and catch · wash cloth · drying off
Arm strengthening	Arm to the front (L1-3, shoulder flexion ≤ 90°)	dnn pose · joint angle vs target · trunk lean / shoulder hike	Arm to the side · elbow · wrist · push-ups · chair-ups
Hand skills	- (Block towers is the first P1 candidate)	-	Block towers · Lego · pick-up sticks · paper clips · poker chips · jars · buttons · laundry
Hand strengthening	-	-	Gripper · putty · squeeze (not camera-verifiable)
Stretching	-	-	All

Why these two beside NHPT

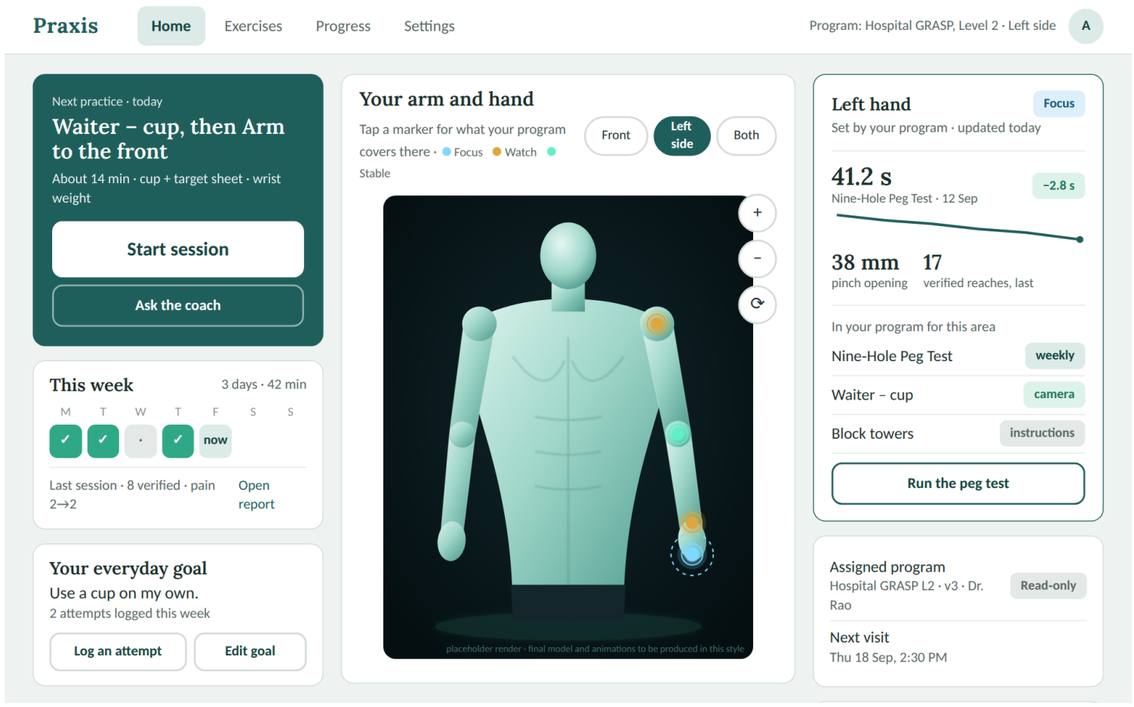
- Waiter - cup reuses the board/object pipeline NHPT already needs; the GRASP target board template is in the instructor manual appendix, so the printed sheet is already a GRASP artifact, and it matches the patient goal “use a cup on my own”.
- Arm to the front is the one pose-tracked exercise the 11 September build plan committed to, and it carries the compensation-detection story (trunk lean, shoulder hiking).
- Each working task exercises a different OpenCV capability, so the judge demo shows three pipelines rather than one pipeline three times.
- Rights note: exercise names appear in the UI; instructions are paraphrased and illustrations are original. Record UBC / Heart & Stroke terms in docs/LICENSES.md before shipping.

2 Information architecture



3 Avatar and reference-panel contract

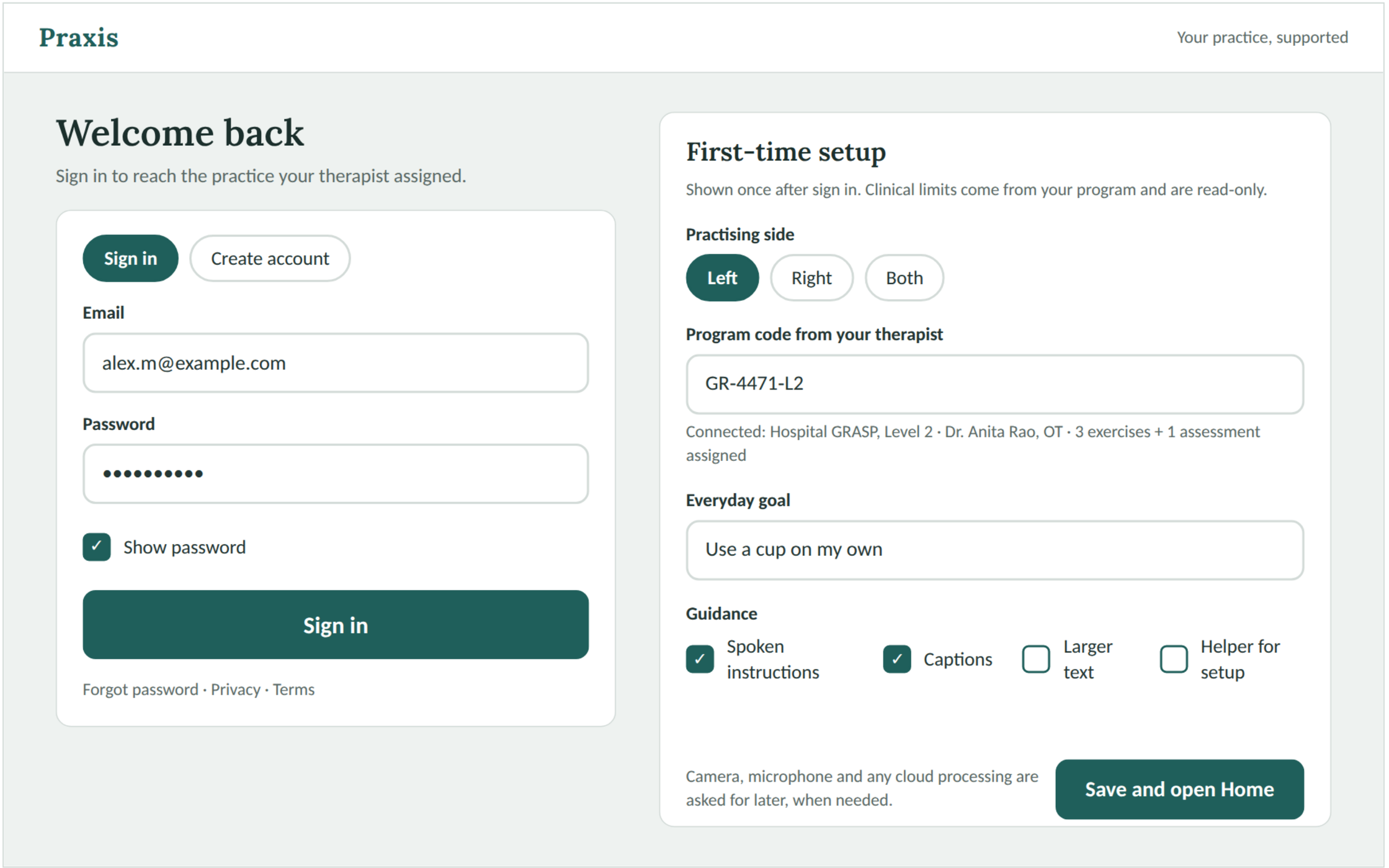
Direction from the supplied references: a smooth aqua mannequin with a featureless head and soft muscle contours on a dark stage, with glowing cue lines and joint dots. Praxis produces its own model and animations in this spirit; reference frames are not reused.



Where it appears	Home centre (body map with region markers); Demonstration; Live session reference panel; Progress region filter (P1). Same figure, same stage, everywhere.
Stage	Always a dark stage card (#050E10 → #0F262B) inside the light interface; unchanged in dark mode. Light theme is the default; dark is a Settings option.
Figure	Seated or standing upper body, front view for hand and reach tasks, side view for shoulder flexion. Practising side mirrored to the patient's own side. No face, no identity.
Cue language	Mint glow = target or ideal path. Amber = watch / current target. Red is reserved for stop. Joint dots: Focus (blue), Watch (amber), Stable (mint), all with a text label nearby.
What it never does	It never scores the patient. Reps are verified from OpenCV events; skeleton similarity to the figure is not an accuracy measure. It never improvises a movement; every clip is clinician-reviewed.
Hand tasks	The verification view is the top-down board (holes, occupancy, next target). The avatar is a pacing reference and can be toggled to Board view.
Data behind markers	Region status comes from the therapist's prescription and camera-derived metrics only. No diagnosis text unless entered by the clinician, shown read-only.
Production plan	Pre-rendered clips per exercise (front and side), plus one idle loop. Live-driven pose mirroring is out of scope for 26 October.

01 Sign in and first-run profile

Three fields to sign in. First-run profile captures side, goal, program code and guidance preferences; clinical limits are read-only from the program. Camera, microphone and cloud consent are asked later, at the point of use.



02 Home (light, default)

The avatar body map is the centre of the page and sets the hierarchy: next practice and week on the left, the selected region's focus panel on the right. Markers show where the program is working, from the prescription and camera metrics. No diagnosis, no composite score.

Praxis | Home | Exercises | Progress | Settings | Program: Hospital GRASP, Level 2 · Left side | A

Next practice · today

Waiter – cup, then Arm to the front

About 14 min · cup + target sheet · wrist weight

Start session

Ask the coach

This week 3 days · 42 min

M T W T F S S

✓ ✓ · ✓ now

Last session · 8 verified · pain 2→2 Open report

Your everyday goal

Use a cup on my own.

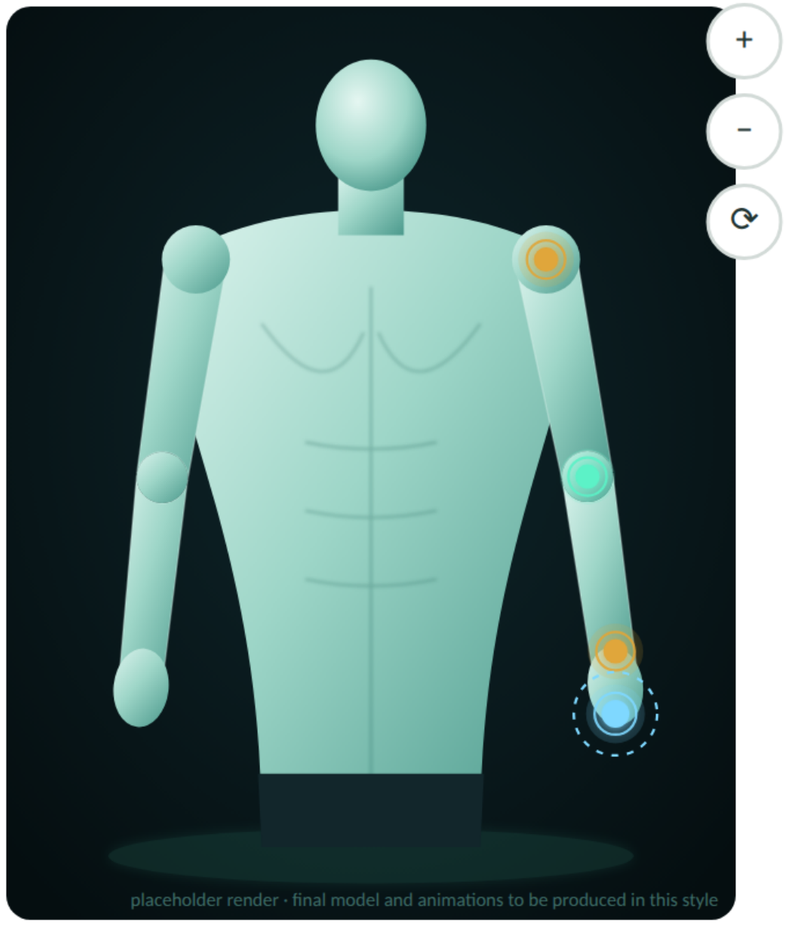
2 attempts logged this week

Log an attempt | Edit goal

Your arm and hand

Tap a marker for what your program covers there · Focus · Watch · Stable

Front | Left side | Both



Left hand Focus

Set by your program · updated today

41.2 s −2.8 s

Nine-Hole Peg Test · 12 Sep

38 mm pinch opening

17 verified reaches, last

In your program for this area

Nine-Hole Peg Test — weekly

Waiter – cup — camera

Block towers — instructions

Run the peg test

Assigned program

Hospital GRASP L2 · v3 · Dr. Rao — Read-only

Next visit

Thu 18 Sep, 2:30 PM

02b Home (dark theme)

Same structure with dark tokens. The avatar stage is unchanged, so the figure reads identically in both themes. Verified/unverified and status colours keep their text labels.

Praxis
Home
Exercises
Progress
Settings
Program: Hospital GRASP, Level 2 · Left side
A

Next practice · today

Waiter – cup, then Arm to the front

About 14 min · cup + target sheet · wrist weight

Start session

Ask the coach

This week

3 days · 42 min

M	T	W	T	F	S	S
✓	✓	·	✓	now		

Last session · 8 verified · pain 2→2

Open report

Your everyday goal

Use a cup on my own.

2 attempts logged this week

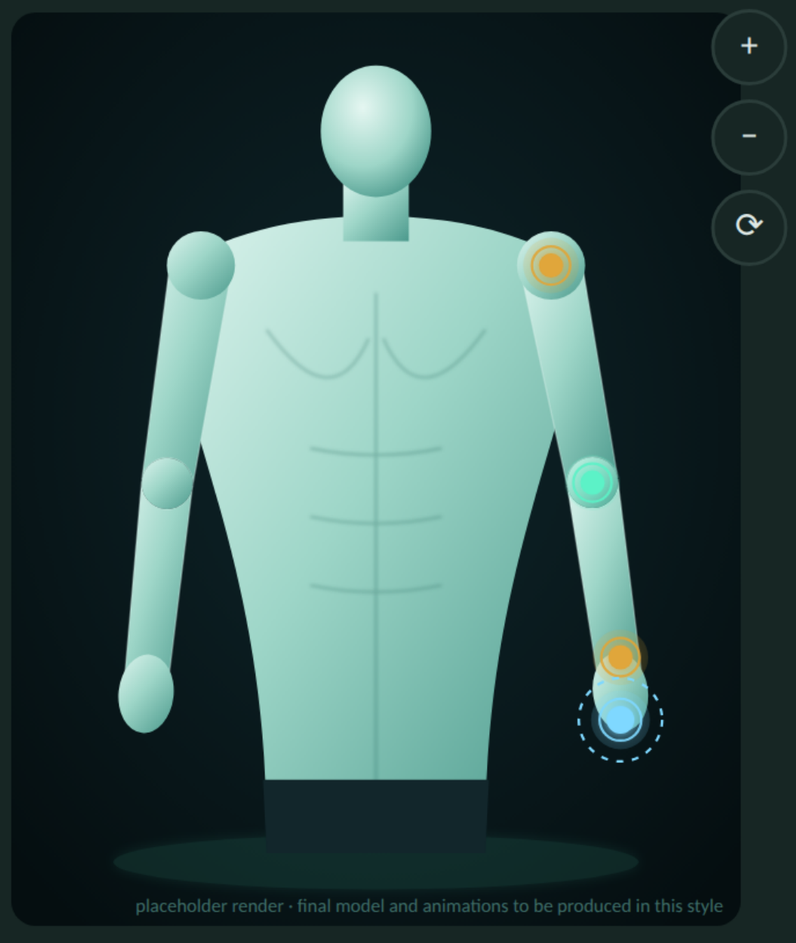
Log an attempt

Edit goal

Your arm and hand

Tap a marker for what your program covers there · Focus · Watch · Stable

Front
Left side
Both



Left hand

Focus

Set by your program · updated today

41.2 s

Nine-Hole Peg Test · 12 Sep

−2.8 s

38 mm

pinch opening

17

verified reaches, last

In your program for this area

Nine-Hole Peg Test	weekly
Waiter – cup	camera
Block towers	instructions

Run the peg test

Assigned program

Hospital GRASP L2 · v3 · Dr. Rao

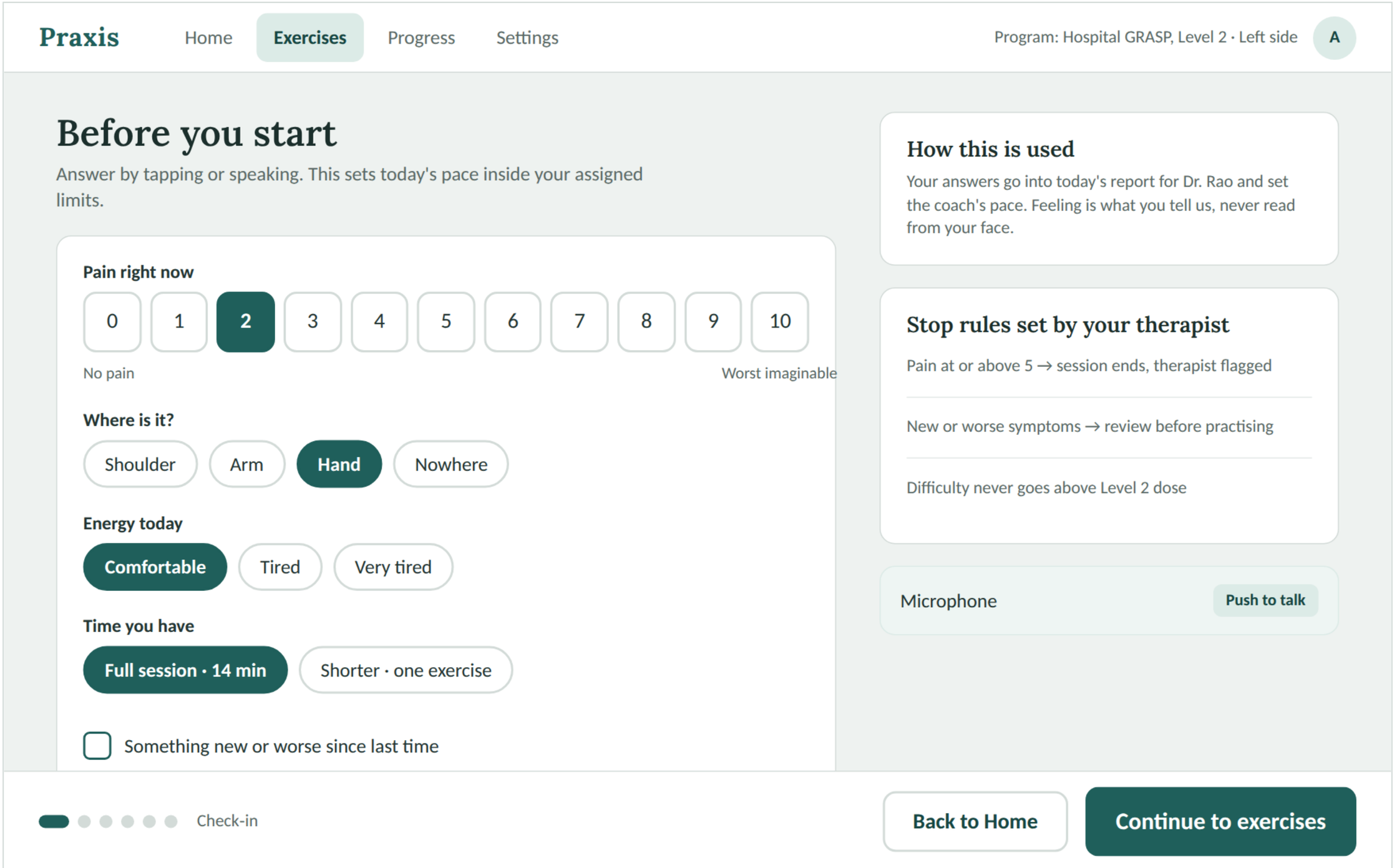
Read-only

Next visit

Thu 18 Sep, 2:30 PM

03 Pre-session check-in

Pain 0-10 with anchors, location, energy, available time and a new-symptoms flag. Therapist stop rules are shown, not inferred. Feeling is self-reported, never read from the face.



04 Choose exercise

Assigned first. Three badges: Assessment, Camera coached, Instructions only. Only validated tasks get Begin setup. Level changes require the clinician.

Praxis Home **Exercises** Progress Settings Program: Hospital GRASP, Level 2 · Left side A

Choose your exercise

Assigned exercises first. Camera coaching is on only where Praxis can verify the task.

Assigned | All GRASP L2 | Equipment I have | **Help me choose**

Standardized assessment · dexterity — **Assessment**

Nine-Hole Peg Test

Pegboard + 9 pegs · printed marker sheet

Timed, one attempt each hand

Timer starts at first peg lift, stops at ninth seated peg.

Begin setup

Coordination · reach and place — **Camera coached**

Waiter – cup

Cup + printed target sheet

2 arcs, then 2 as fast as you can

Camera verifies the cup on each of the five dots.

Begin setup

Arm strengthening · shoulder — **Camera coached**

Arm to the front

Wrist weight 0.5–1 lb

2 sets of 8 · hold 3 · no higher than shoulder

Camera measures arm angle and watches for leaning.

Begin setup

Hand skills · fine motor — **Instructions only**

Block towers

Blocks + stable table

From prescription

Reviewed instructions. Camera coaching not yet validated for this task.

View instructions

Hand strengthening — **Instructions only**

Grip power

5 lb hand gripper

2 sets of 10 · squeeze for 3

Cannot be verified by camera. Logged from your own count.

View instructions

Coordination · both hands — **Instructions only**

Wash cloth

Towel

2 sets of 5

Reviewed instructions. Two-handed twist is not camera-verifiable yet.

View instructions

● ▬ ● ● ● ● Choose

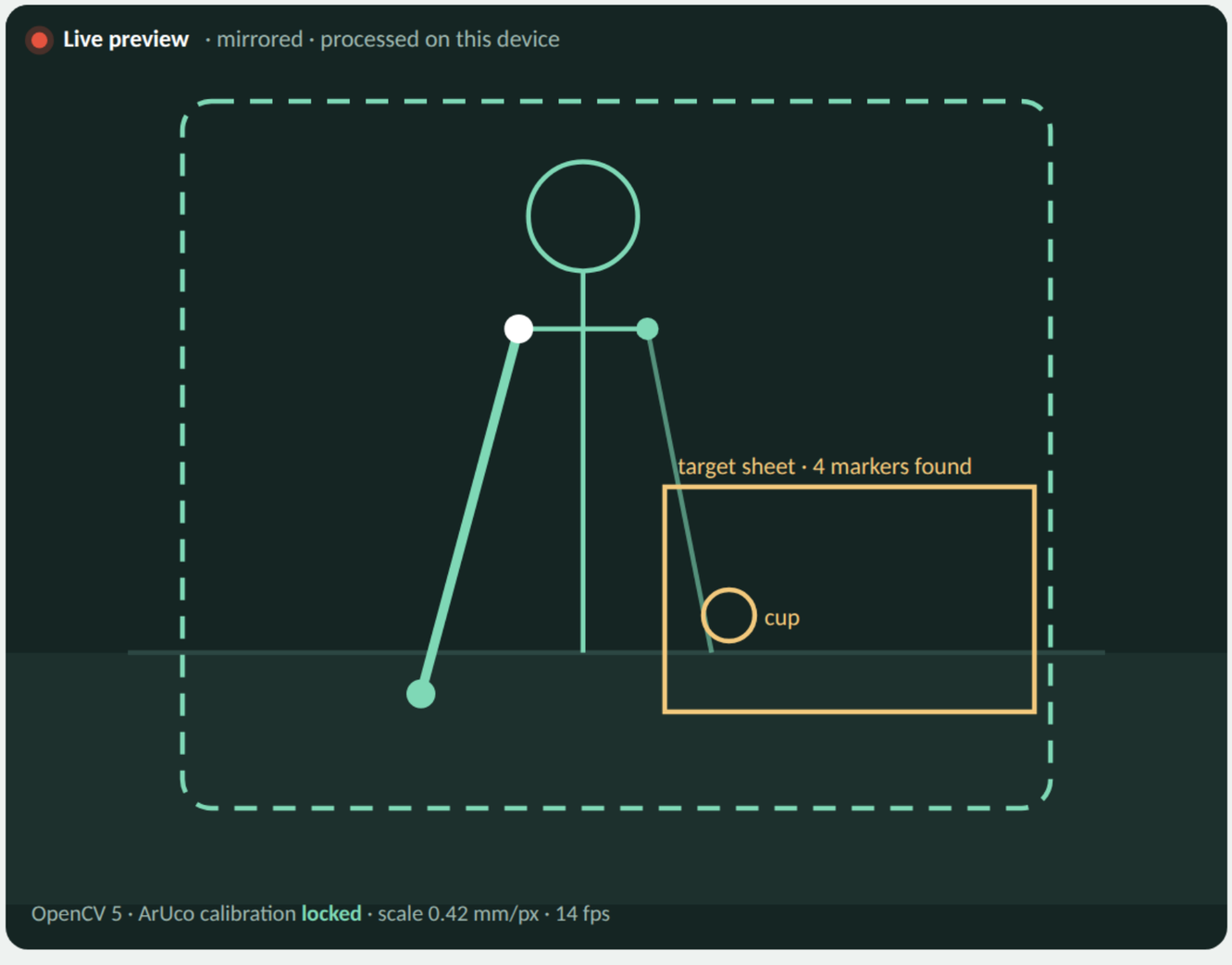
Level changes need your clinician. **Back**

05 Camera setup and calibration

Gated checklist with a fix hint on every failure; Continue stays disabled until required checks pass. Processing location and clip consent are chosen here. ArUco markers required only for board tasks.

Praxis — Waiter – cup · Left side · Setup

Live preview · mirrored · processed on this device



OpenCV 5 · ArUco calibration **locked** · scale 0.42 mm/px · 14 fps

Set up camera and equipment

Continue unlocks when every required check passes.

Pass	Camera permission	Built-in webcam · Change
Pass	One person, left side facing camera	
Pass	Hand and task area in view	
Pass	Cup and target sheet found	All four corner markers visible
Check	Lighting and steady tracking	Move the lamp so your hand is not in shadow

Processing — **On this device** | Cloud, with consent

Video is not stored. Only measurements and events are saved.

Retry | Setup guide | Audio test

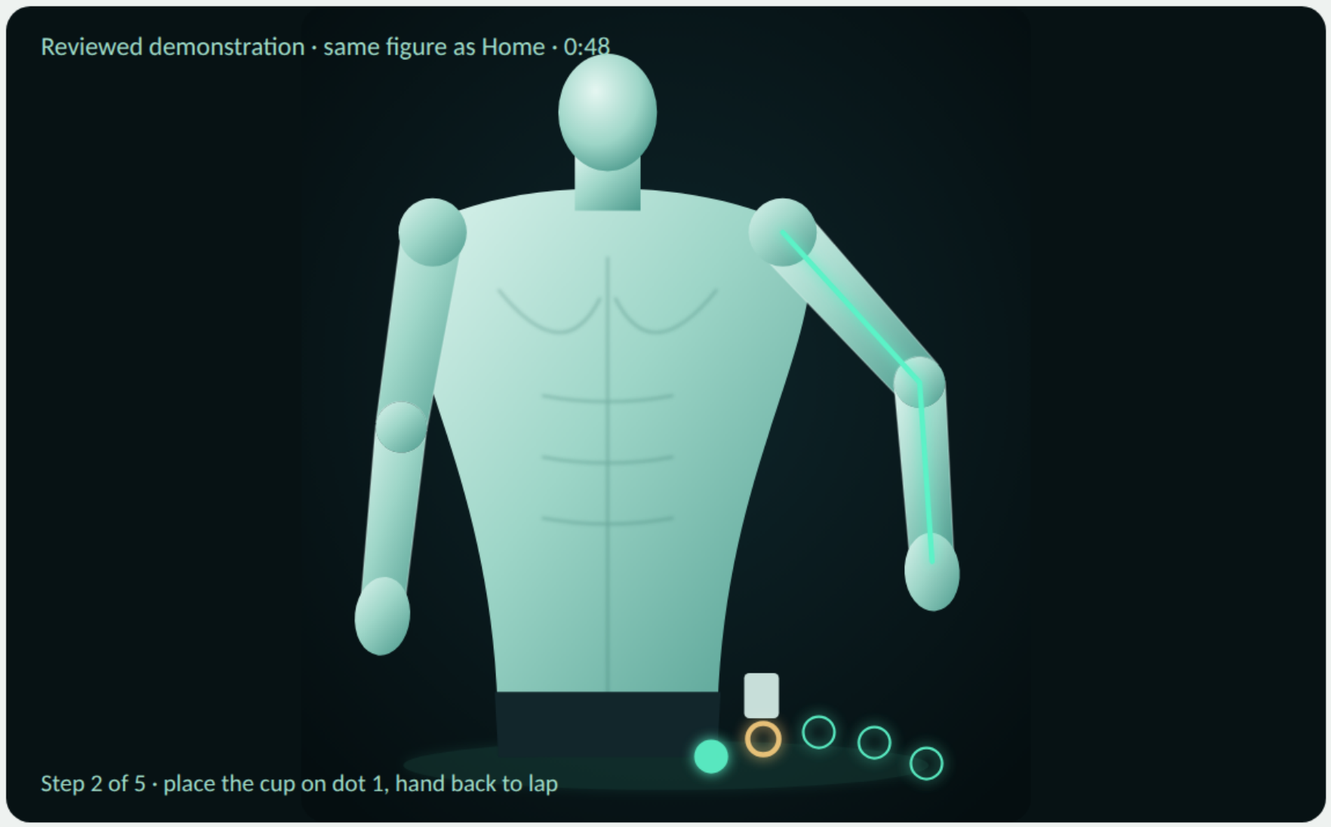
Camera setup

Change exercise | Continue to demonstration

06 Demonstration

The same avatar performs the clinician-reviewed clip on its dark stage: glowing reach path, target dots, current step. Rehearse without scoring is available. Patient confirms readiness before practice starts.

Praxis — Waiter – cup · Left side · Watch first



Play | Replay | Slower | Close-up | Captions

Waiter – cup

Coordination · GRASP Level 2 · left side

1. Cup in your weaker hand, sheet in front of you
2. Place the cup on dot 1, then rest your hand in your lap
3. Reach and move the cup to the next dot, one at a time
4. After dot 5, go back the way you came
5. Repeat twice, then twice as fast as you can

Assigned 2 slow arcs + 2 fast arcs
Watch for shoulder creeping up, leaning to reach
Stop if pain rises or you feel dizzy

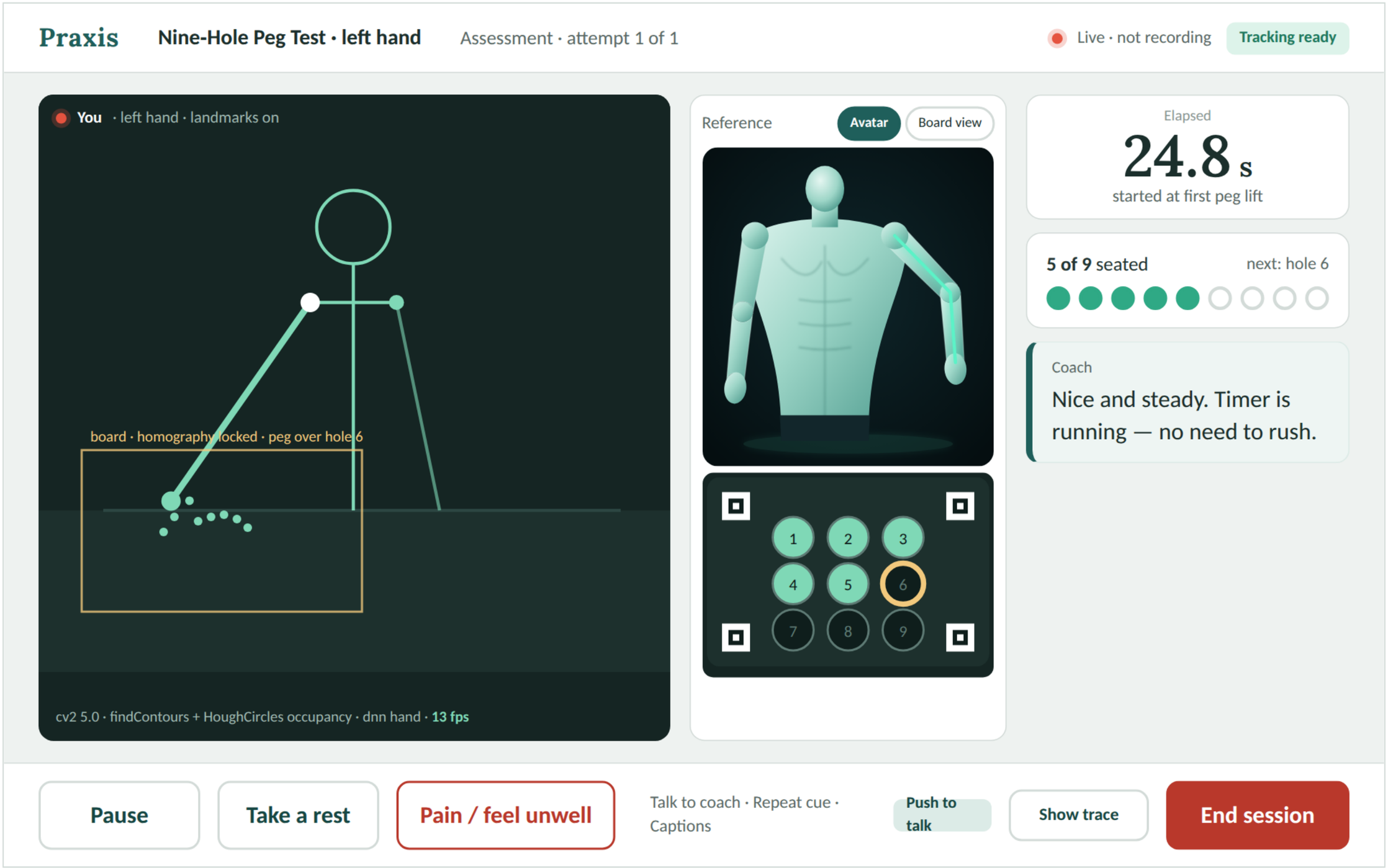
☑ I understand the movement and I'm ready

Demonstration

Ask a question | Rehearse without scoring | Start practice

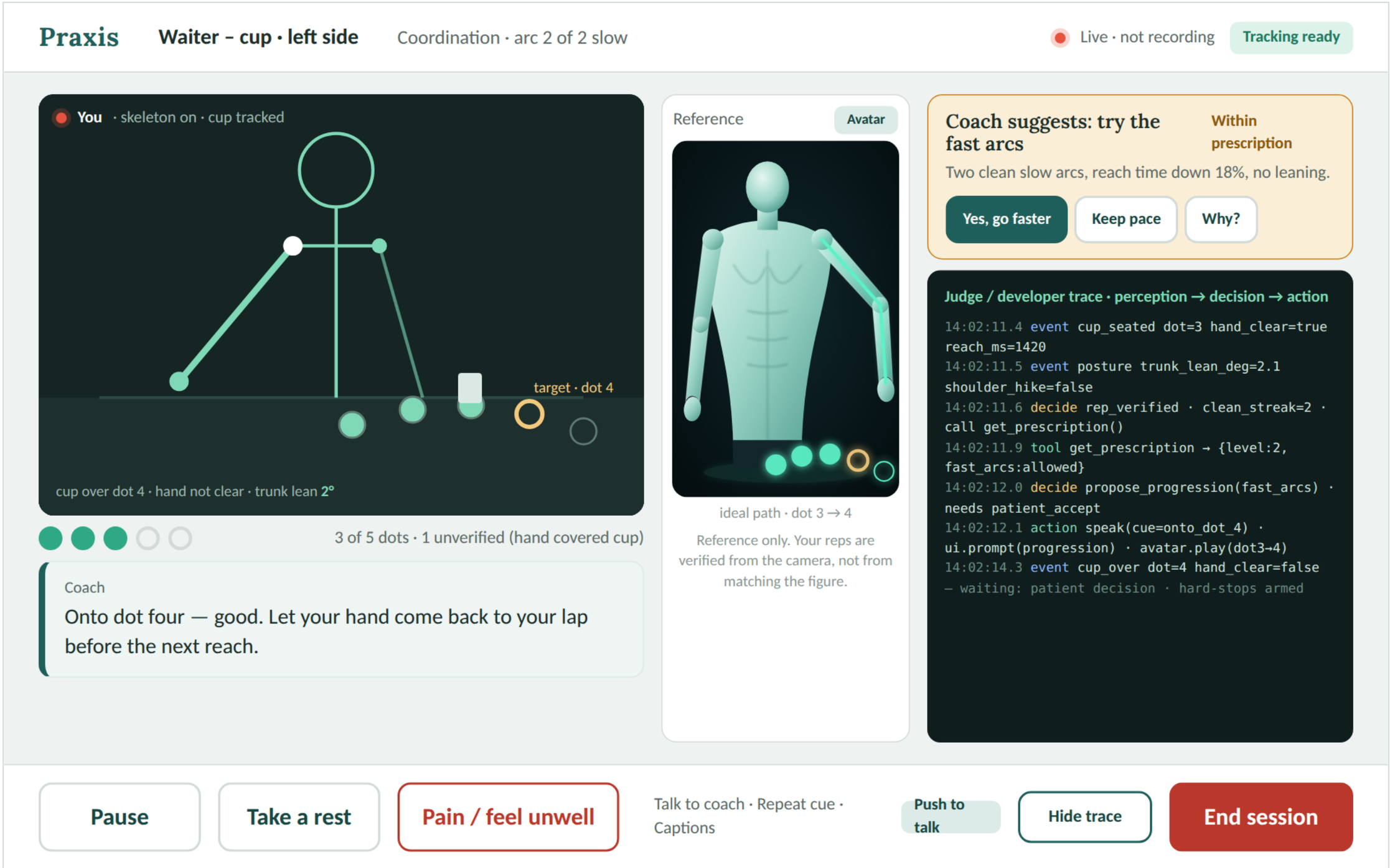
07 Live session - Nine-Hole Peg Test (assessment mode)

Timer starts at first peg lift and stops at the ninth seated peg. The reference panel shows the avatar for pacing and the OpenCV top-down board view for verification; Board view can be made the primary panel.



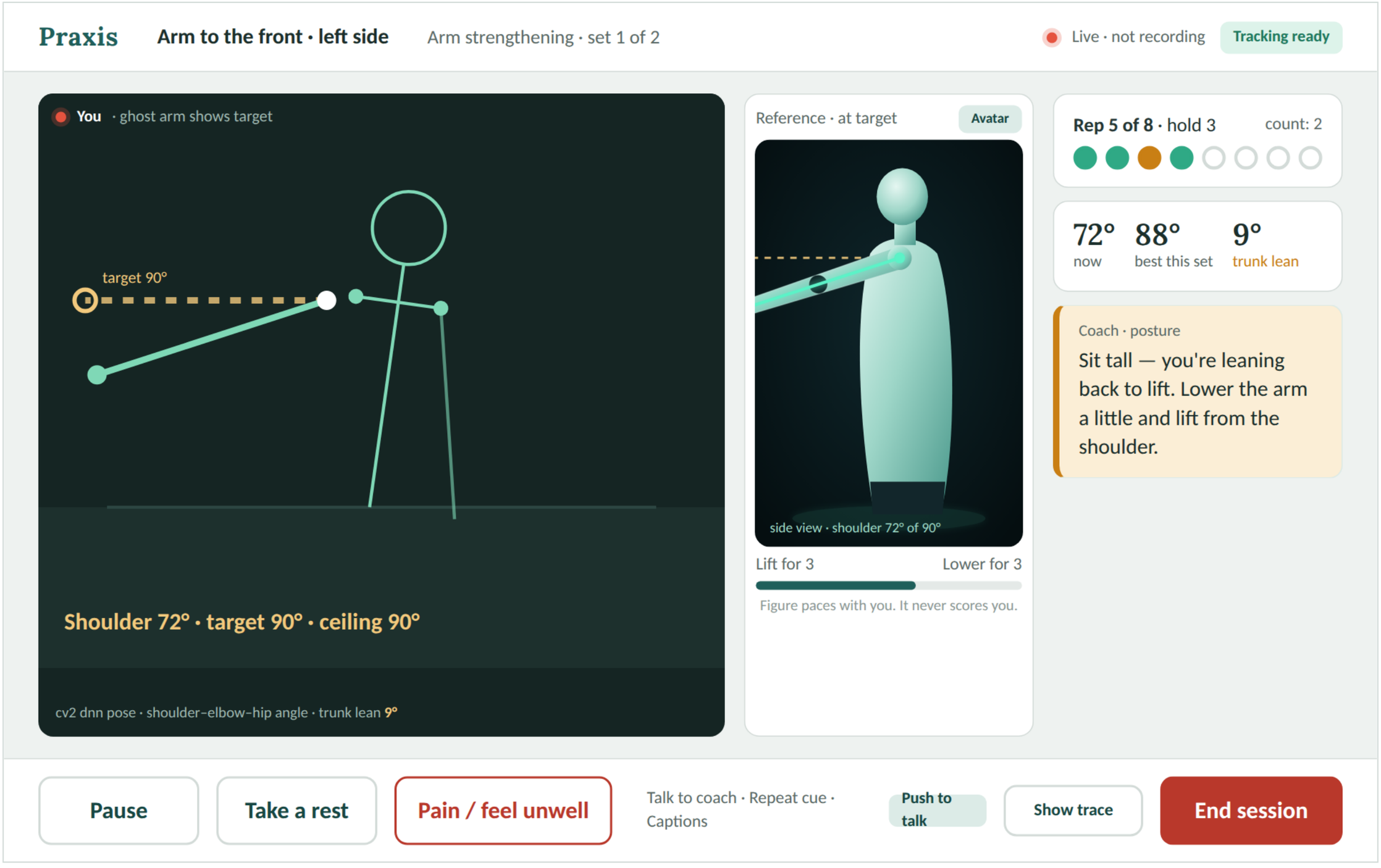
07b Live session - Waiter - cup with trace panel open

Avatar reference shows the ideal path dot 3 → 4 and explicitly never scores the patient. The coach proposes a within-prescription progression; the patient accepts, declines or asks why. The trace now includes the avatar.play call as an action.



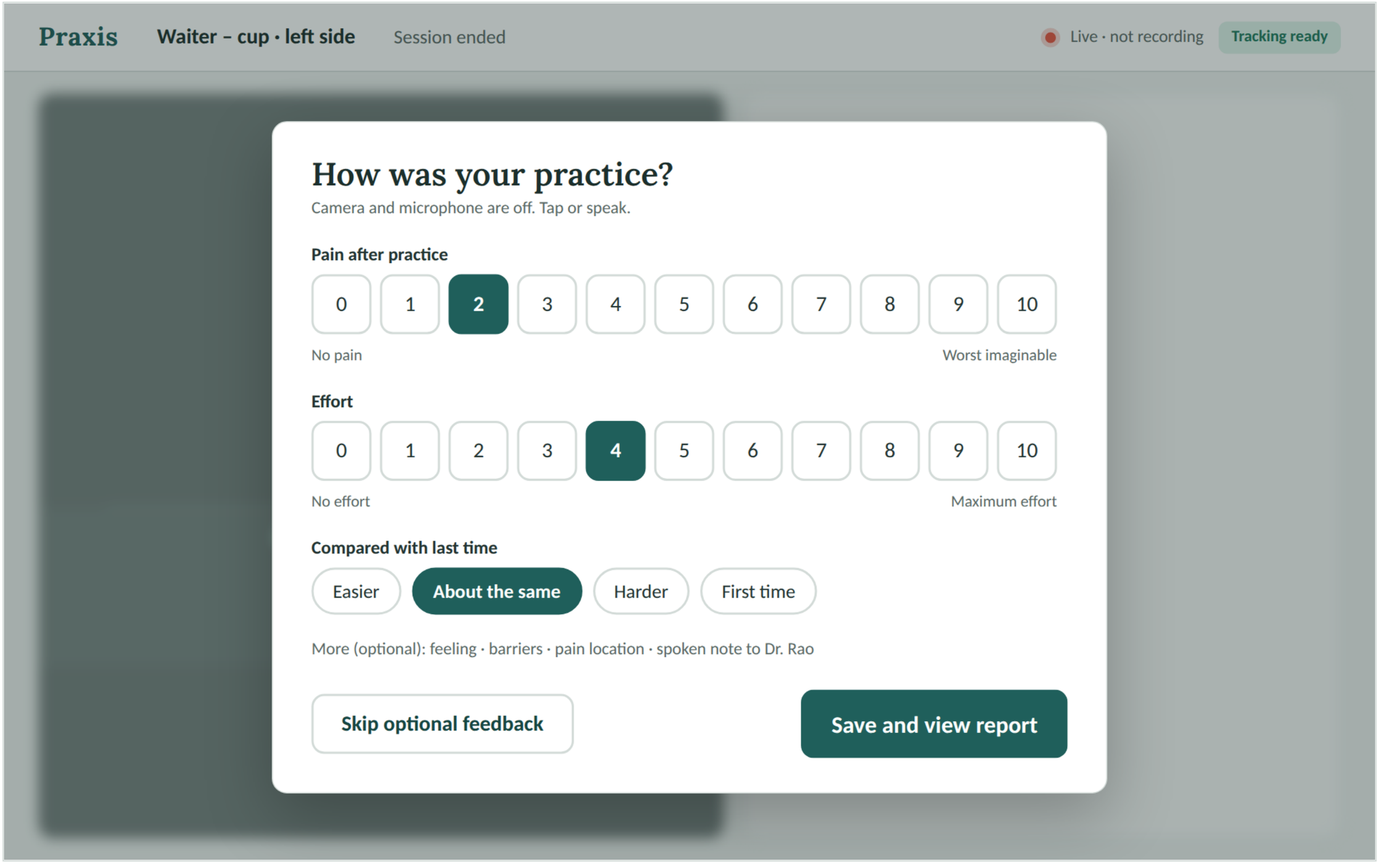
07c Live session - Arm to the front

Side-view avatar holds the target with a mint cue line and dashed 90° ceiling; the camera feed shows the ghost arm and live angle. Trunk lean triggers a posture cue and, if it persists, a within-prescription reduction rather than a push-through.



08 Post-session feedback overlay

Frozen session blurred behind a high-contrast panel. Camera and microphone are off. Pain, effort and comparison; optional detail progressively revealed. Missing means not recorded, never zero.



09 Session report

Camera measurements are separated from what the patient reported. Tracking coverage and unverified attempts are explicit. Coach notes carry confidence labels; next-session changes are proposals; delivery to the clinician is shown as queued until confirmed.

Praxis Home Exercises Progress Settings Program: Hospital GRASP, Level 2 · Left side A

Your practice report

Fri 12 Sep · Waiter – cup + Arm to the front · left side · Complete

Download Sharing Done — back to Home

What the camera measured

11:42	17	3	86%	2
active · rest excluded	verified movements	unverified (hand hid cup)	tracking coverage	posture cues

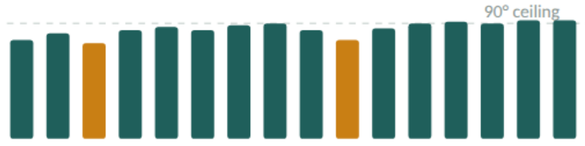
Waiter – cup 2 slow + 2 fast arcs · assigned 2 + 2

Reach time per dot, ms · amber = unverified



Arm to the front 2 × 8 · hold 3 · assigned 2 × 8

Peak angle per rep · ceiling 90°



What you told us

2 → 2 pain before → after

4/10 effort

Felt about the same as last time.

Edit feedback

Coach notes for Dr. Rao

agent-generated · confidence shown

Reach time fell across the slow arcs; fast arcs accepted by patient and completed without drops.	High
Trunk lean appeared in 2 of 16 lifts, late in set 2. Hold time was kept at 3 s; no change made.	Medium
No shoulder hiking observed this session (3 events last session).	High

Next practice

Continue your assigned plan. Fast arcs stay on. No level change — that needs Dr. Rao.

Proposed: keep hold at 3 s, watch leaning — Proposal

Report to Dr. Rao — Queued · not yet delivered

10 Progress and history

NHPT trend and per-task verified counts, filtered to comparable sessions. Limited-tracking sessions are greyed and excluded from comparison. Pain and effort history are patient-reported. No recovery score.

Praxis Home Exercises **Progress** Settings Program: Hospital GRASP, Level 2 · Left side A

Progress and history

Last 4 weeks | Waiter – cup | Left | Compare same setup

Nine-Hole Peg Test

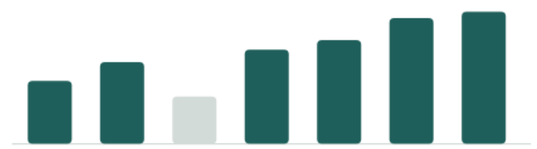
left · lower is better



6 assessments · same board and camera setup · run separately from practice

Waiter – cup

verified reaches per session



7 sessions · grey = limited tracking, not comparable

6 → 8 verified · same task, side and setup

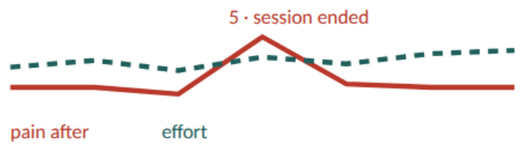
Practice

19/22 planned sessions done

3.4 h active time, 4 weeks

Pain and effort

What you reported, per session



Session list

12 Sep	Waiter – cup + Arm to the front	Complete	Usable tracking	Open
12 Sep	Nine-Hole Peg Test · 41.2 s	Assessment	Usable tracking	Open
10 Sep	Waiter – cup	Complete	Limited tracking	Open
8 Sep	Arm to the front	Ended early · pain 5	Usable tracking	Open

Everyday goal log

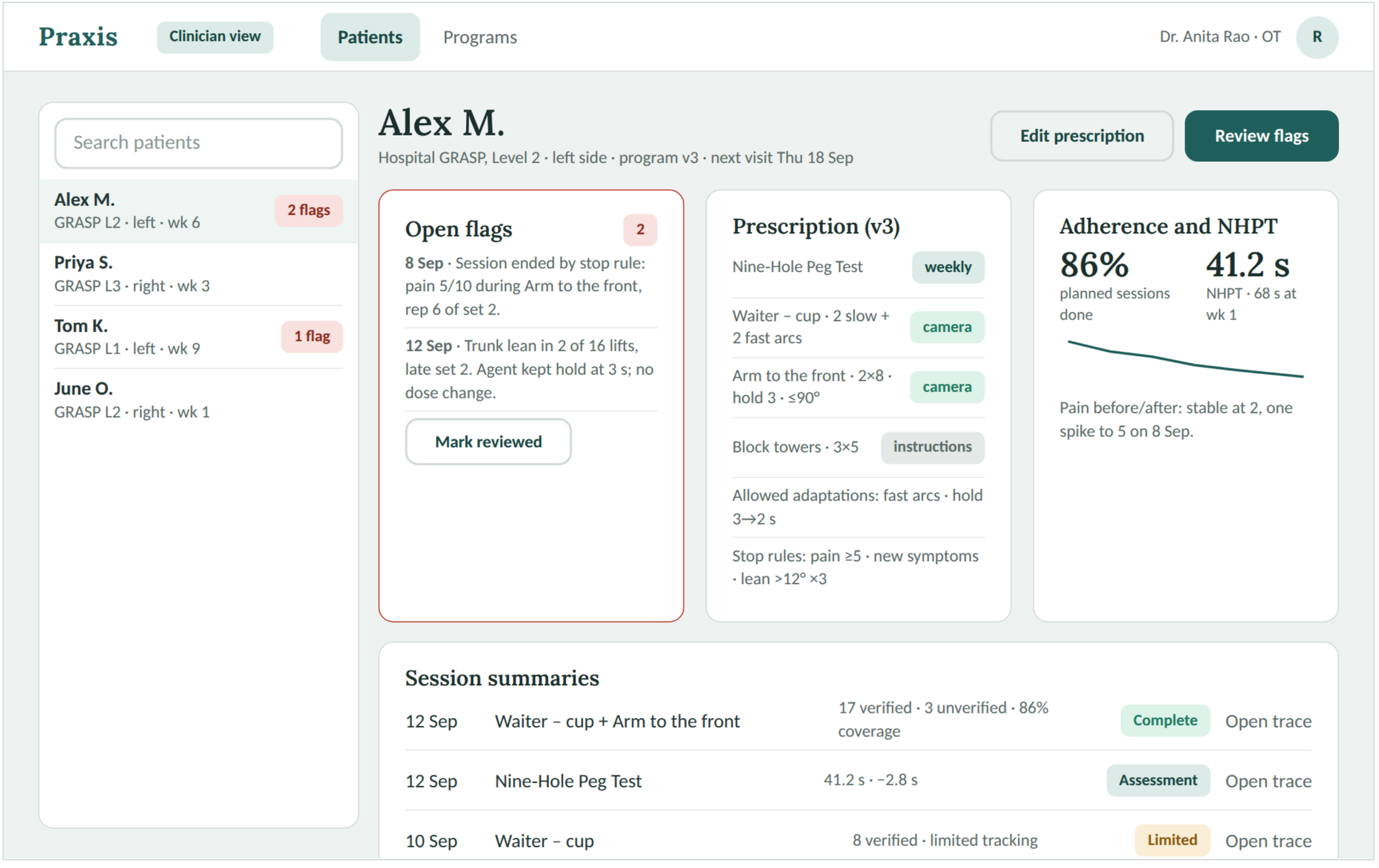
Used a cup with my left hand — 6 times this month

Log an attempt

No recovery score is calculated. Different tasks are never combined.

11 Therapist dashboard (read-only for the competition build)

Open flags from stop rules and compensation, the versioned prescription with allowed adaptations and stop rules, adherence and NHPT, and session summaries with a link to each decision trace.



5 Design rules applied to every screen

Tap targets ≥ 56 px; one full-width primary action per flow screen	one-handed use with the stronger hand; tremor tolerance
Body text ≥ 18 px, ≥ 24 px for coach cues in session; captions for all speech	aphasia, hearing, distance from the screen
Every action reachable by voice (push-to-talk) and keyboard	hands are occupied during exercises
Pause · Rest · Pain / feel unwell · End fixed on every in-session screen	human control criterion of the Agentic Vision award
'Live · not recording' indicator whenever the camera is active; process-only by default	responsible-use evidence
Verified vs unverified counts everywhere; no composite accuracy or recovery score	never overstate what a webcam can measure
Clinician prescribes; the coach proposes within it; stop rules override the coach in code	structural guardrails, not prompt guardrails
Judge/developer trace behind a toggle on the live screen	perception → decision → action evidence without burdening the patient
Colour never carries meaning alone: mint = verified, amber = cue/unverified, red = stop, always with text	WCAG 2.1 AA
Palette #1F5F5B teal, #2FA985 mint, #C97F14 amber, #B9382B red on #EEF2F0; Lora headings, Carlito UI text	calm, high-contrast, distinct from generic templates
Light theme by default; dark theme in Settings; the avatar stage is dark in both	one consistent figure across Home, Demonstration and Session